

OmniBrain (S.A.R.A.) - Comprehensive Technical Documentation

Section 1: Executive Summary & Core Architecture

OmniBrain is a highly advanced, autonomous AI Operating System built for extreme speed and precision. At its core is S.A.R.A. (Smart Autonomous Responsive Assistant). The system is containerized using Docker, orchestrating multiple microservices:

- Frontend: Next.js 15, React 19, Tailwind CSS, providing a real-time Chat UI.
- Backend: FastAPI (Python 3.12) utilizing asynchronous endpoints for zero-blocking operations.
- Database: PostgreSQL 16 equipped with 'pgvector' for high-dimensional memory embeddings.
- Message Broker & Cache: Redis 7 for instant response caching and background task queueing (Arq).
- AI Model: Integrated with Gemini (gemini-3.5-flash-lite / 3.8-pro) via a dynamic ModelRouter.

Section 2: The Cognitive Brain Engine (A.I. Pipeline)

The 'Brain' of Sara operates through an ultra-fast, multi-stage pipeline to process user intent:

- IntentEngine: Uses a sub-millisecond heuristic fast-path. It analyzes the raw user query and classifies it as either 'Conversational' (greeting/QA) or 'Actionable' (requires tool execution).
- ContextEngine: Rapidly queries PostgreSQL to assemble the User Profile, active connected integrations, recent background tasks, and long-term vector memories into a dynamic context snippet.
- DAGPlanner (Directed Acyclic Graph): If a task is actionable, the Planner breaks it down into a logical sequence of steps. For example, 'Find latest email and summarize' becomes [Step 1: gmail.read] -> [Step 2: internal.summarize].
- ResultSynthesizer: Once the background worker completes the DAG steps, the Synthesizer compiles the raw JSON outputs into a natural, spoken-language report for Sara to speak.

Section 3: Integrations, Tools, and Connectors

Sara is equipped with real-world tool connectors allowing her to read, write, and execute external actions:

- Google Mail (Gmail): Full read/write access. Sara can search mailboxes, draft new emails, and send emails directly using OAuth2 tokens.
- Google Calendar: Capability to read upcoming events and schedule new meetings seamlessly.
- Google Drive: Search capabilities across the entire Drive workspace.
- Google Sheets: Read rows, update specific cells, and append new rows to spreadsheets.
- Web Search Agent: Integrated with DuckDuckGo Search API to fetch live, real-time internet data. If a user asks for 'latest news', Sara scrapes the web dynamically.

Section 4: Voice Mode & Human-Like Personality

Sara's interaction layer has been completely overhauled to mimic 'ChatGPT Voice Mode' behavior.

1. Text-to-Speech (TTS): Utilizing Microsoft Edge Neural TTS (edge-tts). The default Hindi/Hinglish voice is

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'hi-IN-SwaraNeural'. We applied custom acoustic tuning (Pitch: -2Hz, Rate: +5%) to remove robotic artifacts, making her sound relaxed and deeply human.

2. System Prompt Engineering: Sara operates under strict personality constraints. She is instructed to act warm, empathetic, and professional. She uses natural conversational Indian fillers (e.g., 'Haan ji', 'Dekhiye', 'Bataiye') instead of robotic, bullet-point lists.

Section 5: Extreme Performance Optimizations

To achieve 'Flash Speed' (responses under 1 second), several critical low-level optimizations were implemented:

- Pre-Warmed Singletons: Previously, engines (Router, Context, Planner) were initialized on every request, costing ~500ms. They are now instantiated once at startup and shared across the application.
- Redis Response Caching: Common conversational queries are hashed and cached in Redis with a 5-minute TTL. Repeated questions return in literally 0ms.
- SSE Streaming (Server-Sent Events): Replaced synchronous waiting with an async generator. Sara's response streams word-by-word to the Next.js UI, giving the illusion of instant generation.

Section 6: Security & Concurrency

- Authentication: Implemented secure JWT (JSON Web Tokens) with a 30-minute expiry for API security.
- Safe Task Execution: Destructive tasks (e.g., deleting files, sending emails) are routed to a 'WAITING_APPROVAL' state, ensuring human-in-the-loop validation.
- Arq Redis Workers: Tool execution happens entirely in the background. The main API thread is never blocked, allowing OmniBrain to handle hundreds of concurrent requests.

Section 7: Upcoming / Work-in-Progress

OmniBrain is continuously evolving. The next immediate milestone is the Windows Host Agent (Local PC Control). Because OmniBrain runs safely inside a Docker Linux Sandbox, it cannot directly control the host Windows PC. We are designing a bridge script ('host_agent.py') that will allow Sara to break out of the sandbox to open local files, type on the keyboard, and execute native Windows CMD operations.